TASK 3 · CLUSTERING

Which clusters are biologically meaningful?

K-means, hierarchical clustering and Leiden

20,000
live PBMCs

37
biological markers

20
donors · 1,000 cells each

A good geometric score does not establish a biological cell type.

Our analysis: notebook 03_clustering, run on 7 September 2026. PBMC = peripheral blood mononuclear cell.

Same data, three families of methods

Preprocessing

1,000 PBMCs per donor, seed 42

↓

$\operatorname{arcsinh}(x/5)$

↓

Z-standardisation per marker

↓

X: 20,000 cells $\times$ 37 markers

44 configurations

- K-means: 7 values of k, 10 starts each.
- Hierarchical: single, average, complete and Ward; 7 values of k each.
- Leiden: 3 neighbour counts $\times$ 3 resolutions; weighted graph built from X.

$k \in \{2, 3, 4, 6, 8, 10, 12\}$
$n \in \{10, 30, 60\}$; $r \in \{0.3, 0.7, 1.2\}$

Selection per method: lowest Davies–Bouldin index among solutions with stability ≥ 0.6.

Stability: 10 repeats retaining 80 % of cells per donor; refit scaling and clustering. Details on backup slide 10.

Selection produces very different solutions

Method	Parameters	Clusters	DB ↓	Stability	Largest cluster
K-means	$k = 4$	4	2.363	0.995	37.780 %
Single	$k = 2$	2	0.264	1.000	99.995 %
Average	$k = 2$	2	0.894	0.602	99.975 %
Complete	$k = 2$	2	0.948	0.617	99.960 %
Ward	$k = 6$	6	2.522	0.611	38.220 %
Leiden	$n = 30, r = 1.2$	18	2.361	0.683	17.940 %

K-means and Leiden: similar DB scores, different granularity and stability.

Check cluster sizes: single, average and complete each put over 99.9 % of cells in one cluster.

DB = Davies–Bouldin index, lower is better. Stability = unweighted mean of per-cluster median Jaccards. Our notebook results.

Why the best DB score is misleading here

Single linkage, $k = 2$

19,999 cells in C0

1 cell in C1

99.995 % of cells in one cluster

Yet the metrics look excellent

DB: **0.264**

Stability: **1.000**

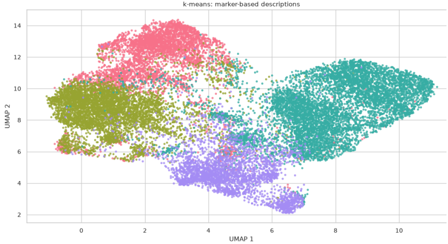
Separating an isolated event can be geometrically favourable and reproducible under resampling.

This solution mainly isolates one outlier; it does not answer the cell-type question.

The bar shows the large group's share; the single cell would not be visible at this scale. Our results; DB definition: scikit-learn.

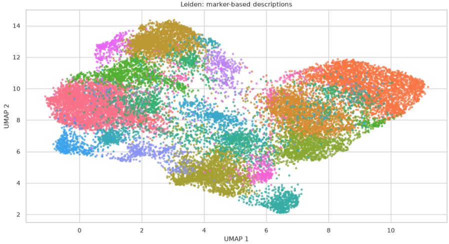
K-means forms broad groups; Leiden splits them further

K-means · 4 clusters



3,148–7,556 cells per cluster
Mean stability: **0.995**

Leiden · 18 clusters

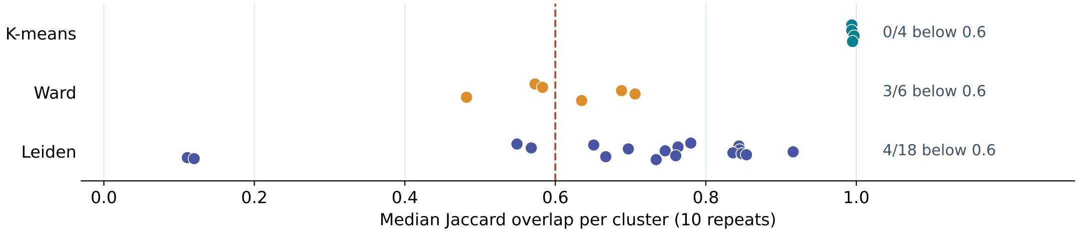


223–3,588 cells per cluster
Mean stability: **0.683**

The finer partition provides candidate subgroups whose stability needs to be assessed individually.

Original notebook UMAPs, with legends hidden for comparison. Same embedding; colours are local to each method. Clustering and scoring use all 37 markers, not UMAP.

The mean hides unstable clusters



Leiden: 4 of 18 below 0.6
C15 and C16 reach only 0.111 and 0.120.

Ward: 3 of 6 below 0.6
The configuration remains eligible with a mean of 0.611.

The 0.6 threshold applies to the configuration's mean, not to every individual cluster.

Each point represents a cluster of the selected solution. Saved median Jaccards; the dashed line helps compare individual values.

Marker profiles suggest biological hypotheses

Hypothesis	K-means	Leiden	Interpretation
B-cell-like	C0: CD19↑, HLA-DR↑, CD3↓	C3: CD19↑, HLA-DR↑, CD3↓	Similar relative profiles
NK-cell-like	C3: NKp46↑, CD94↑	C4: NKp46↑, CD122↑, CD3↓	Candidates; check further markers
Unclear	C1: CD34↑, CD33↑; C2: CD127↑	Further subgroups	Profiles alone do not support an assignment

Arrows describe relative cluster medians.

They are not positive/negative gates or evidence of coexpression within individual cells.

These profiles support cell-type hypotheses, but do not replace validated annotations.

Our preliminary interpretation of notebook profiles. Biological context: Horowitz et al. (2013), doi:10.1126/scitranslmed.3006702. Similar profiles do not establish overlap in cell membership.

Our assessment: stable groups, unvalidated cell types

K-means as a broad starting point

Four highly stable groups; plausible marker profiles for some clusters.

Leiden for finer hypotheses

More subgroups, including several unstable clusters. No established biological advantage overall.

To assess biological relevance

- Check marker combinations and coexpression.
- Inspect distributions by donor and acquisition date.
- Evaluate fresh cell samples, especially for small groups.

The selected single, average and complete solutions are highly degenerate; Ward is less stable.

Combine quantitative assessment with marker interpretation. A clear biological winner is not yet established.

Assessment of selected solutions on this sample, not a general ranking of the algorithms.

Backup: what changes from 200 to 1,000 cells per donor?

Method	200 cells/donor	1,000 cells/donor
K-means	$k = 4$; stability 0.973	$k = 4$; stability 0.995
Single / Average / Complete	$k = 4$ each	$k = 2$ each
Ward	no eligible configuration	$k = 6$; stability 0.611
Leiden	$n = 30$, $r = 0.7$; 8 clusters	$n = 30$, $r = 1.2$; 18 clusters

Two changes at the same time:

- Five times as many cells: 4,000 $\rightarrow$ 20,000 in total.
- Added $k = 2$ and $k = 3$ to the grid for distance-based methods.

The differences cannot be attributed to the larger sample alone.

200-cell reference: verified rerun with threshold 0.6, not the earlier stale notebook outputs. Sources: both investigation reports in the repository.

Backup: how is stability calculated?

In repeat b, we compare the same retained cell IDs R_b:

$$J_{c,b} = \max_j \frac{|(C_c \cap R_b) \cap D_{j,b}|}{|(C_c \cap R_b) \cup D_{j,b}|}, \qquad S = \frac{1}{K} \sum_{c=1}^{K} \mathrm{median}_b\, J_{c,b}.$$

C_c: original cluster; $D_{j,b}$: new cluster on retained cells.

- 10 repeats; retain 80 % within each donor, without replacement.
- Refit scaling and clustering; method parameters stay fixed.
- Absent clusters: NaN. At least 6 assessable repeats in the current run.
- No minimum size; a singleton can be stable when retained.

This measures robustness within the initial sample, not generalisation to new donors.

Our notebook implementation; reference assignments are not ground truth.

Backup: sources and reproducibility

Our results

`notebooks/03_clustering.ipynb`, run on 7 September 2026.

20,000 cells, seed 42, arcsinh cofactor 5, Z-standardisation; 44 + 440 fits.

Tables and UMAPs exported directly from saved notebook outputs.

Dataset and biological context

Horowitz et al. (2013). Genetic and environmental determinants of human NK cell diversity revealed by mass cytometry. *Science Translational Medicine*.

doi:10.1126/scitranslmed.3006702

Arvaniti & Claassen (2017). Sensitive detection of rare disease-associated cell subsets via representation learning. *Nature Communications*.

doi:10.1038/ncomms14825

Method documentation

scikit-learn: Davies–Bouldin index   SciPy: linkage   Scanpy: Leiden